Low-Rank and Sparse Representation Meet Deep Unfolding: A New Interpretable Network for Hyperspectral Change Detection

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***Abstract*— Hyperspectral image change detection (HSI-CD) is a technique that intelligently checks the changed details in bitemporal hyperspectral images (Bi-HSIs). Deep learning (DL), with the ability to model nonlinear changing features, has achieved promising results in HSI-CD, but the feature mining mechanism is unclear and the architecture design lacks transparency in such DL models. To alleviate this problem, this article proposes a new low-rank and sparse representation-based deep unfolding network (LRSRNet) for HSI-CD. For feature mining mechanism, the LRSRNet adopts a low-rank and sparse subnetwork (LRSnet) and a change detection subnetwork (CDnet). The former is responsible for extracting low-rank features with valuable information and suppressing sparse features containing interference information, while the latter aims to obtain change information from low-rank features. For architecture design, the LRSnet formulates the HSI as a low-rank estimation, sparse estimation, and hyperspectral reconstruction in a low-rank and sparse model, and iteratively optimizes and updates the above subproblems through deep networks. A new CDnet is designed as a concise convolutional architecture to extract change information from representative Bi-HSIs features. Experiments on three real datasets demonstrate the performance superiority of the proposed LRSRNet method over nine model-driven, data-driven, and model–data joint-driven HSI-CD algorithms in both qualitative and quantitative evaluations. The proposed LRSRNet is available online: https://github.com/chengle-zhou/LRSRNet**

***Index Terms*— Bitemporal hyperspectral images (Bi-HSIs), change detection, deep unfolding, low-rank and sparse representation.**

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I. Introduction

HYPERSPECTRAL image (HSI) is a fine-spectral Earth observation technique that provides diagnostic information on ground covers from visible light to shortwave infrared [1], [2], [3]. HSI change detection (HSI-CD) is a technical means to utilize HSIs of the same scene at different times [e.g., bitemporal HSIs (Bi-HSIs)] to realize the perception of pixel, region, and scene change information, which has been widely used in precision agriculture [4], military reconnaissance [5], and disaster assessment [6].

The earliest HSI-CD methods can be roughly divided into three categories, namely, algebra-based methods, transformation-based methods, and classification-based methods [7]. Algebraic methods often use image difference information (e.g., spectral difference, absolute distance, and ratio) and image regression to obtain ground changes in Bi-HSIs scenes. For example, change vector analysis (CVA) determines the degree of coverage change based on the magnitude and direction of the change vector of Bi-HSIs [8], [9]. However, CVA cannot take into account the spectral variability caused by solar altitude, atmospheric conditions, and soil wetlands, resulting in serious pseudo-change pixels in the results (i.e., a high false alarm rate). The basic design of transformation-based methods is to project the original data to find a feature space that can linearly distinguish between changing and nonchanging information [10]. Celik [11] employed principal component analysis (PCA) to obtain low-dimensional representative features of the original HSIs and used k-means clustering to further obtain the change information in the scene. Nielsen [12] introduced an iterative reweighted MAD method based on canonical correlation analysis between Bi-HSIs. Wu et al. [13] proposed a slow feature analysis to capture the significance of changes between Bi-HSIs. Classification-based methods include postclassification and direct classification. The former respectively classifies the bitemporal observation images and obtains change information through class difference analysis. The latter uses a classifier to perform binary classification on its difference features to obtain change results. Classic methods include support vector machine (SVM) [14], maximum likelihood algorithm [15], and k-nearest neighbor (k-NN) [16]. The above traditional HSI-CD methods can be summarized as empirical mathematical models (i.e., model-driven), which have the advantage of clearly defining

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the feature extraction mechanism and change detection rules in Bi-HSIs. That is, the mechanism is simple and clear. However, the above methods suffer from noise interference in practical applications, and there are problems such as insufficient feature utilization and difficulty in detecting subtle changes.

Recently, data-driven deep learning (DL) methods have achieved promising performance in the field of HSI-CD, thanks to the ability of deep networks to capture nonlinear changing characteristics [17], [18], [19], [20]. Representative deep network architectures include recurrent neural networks (RNNs) [21], convolutional neural networks (CNNs) [22], and transformers [23]. Mou et al. [21] introduced a recurrent CNN for change detection in remote sensing images to learn a joint spectral–spatial–temporal feature representation framework. Chen et al. [24] proposed a deep Siamese convolutional multilayer RNN to obtain change information in dual-temporal images. Shi et al. [25] designed a multipath convolutional RNN to capture the multiscale temporal–spatial–spectral variation characteristics in Bi-HSIs. The above RNNs architectures all use long short-term memory (LSTM) to associate the changing details of the dual-phase image scenes [26]. Du et al. [22] and Wang et al. [27] designed CNN-based HSI-CD architectures from the perspective of deep representation of slow features and endmember abundance, respectively. Ou et al. [28] proposed a CNN framework with compact band weighting and multiscale spatial attention for HSI-CD. A multidecision joint alignment framework using CNNs are designed for HSI-CD in [29]. Since CNN, which only has the ability to summarize local features, has difficulty capturing global dependencies, the transformer architecture came into being. Li et al. [23] proposed a cross-band 2-D self-attention network to preserve the spectral characteristics in HSI-CD task. Wang et al. [30] developed a triple-branch transformer network based on parent and sibling attention for HSI-CD. Xiao et al. [31] proposed a novel selective transformer architecture that effectively addresses the bottleneck in feature extraction. Besides, convolutional transformer-based network architectures have also been widely investigated in HSI-CD applications [32], [33]. Although the above data-driven methods have achieved promising detection results in the HSI-CD task, there are three aspects that deserve further attention.

1) *Architecture Design:* Deep networks all use basic units such as convolutional layers and fully connected layers to build network architectures, ignoring experience-guided modeling.
2) *Feature Mining:* It relies on complex deep networks to abstract and represent the temporal–spatial–spectral features in Bi-HSIs gradually, ignoring the domain knowledge of image processing.
3) *Sample Desire:* Detection performance relies on iterative training of a large amount of labeled data, and the performance will be unsatisfactory when samples are limited.

More recently, the model–data joint-driven approaches have been favored by scholars in the field of HSI processing due to the transparency of the model architecture and its lightweight characteristics [34], [35], [36]. For instance, Li et al. proposed a series of interpretable deep unfolding network schemes for anomaly detection problem [37], [38]. Nie et al. [39] proposed a deep unfolding network representation model for HSI classification task. Xiao et al. [40] introduced an efficient diffusion probabilistic model for super-resolution task, which shows remarkable reconstruction performance. Besides, Zhang et al. [41] introduced a spatial–spectral dual-image unfolding network for multispectral and hyperspectral fusion. In the HSI-CD task, some scholars have designed DL networks based on the dictionary learning mechanism for HSI-CD, which on the one hand enhances the transparency of the network structure and on the other hand improves the detection performance of the dictionary learning algorithm. Zhao et al. [42] designed a dictionary learning-guided deep interpretability network for HSI-CD. The fundamental concept is to assume that Bi-HSIs share an overcomplete dictionary and thus define scene attributes (i.e., changed or unchanged) based on the sparse difference. Qu et al. [43] introduced the multiscale strategy into the dictionary learning network and proposed a multiscale convolutional sparse coding-guided deep interpretable network for HSI-CD. The above literature provides successful cases and solutions of the model–data joint-driven paradigm in the field of HSI processing.

In this article, we explore the unfolding network paradigm of coupling low-rank and sparse representation model with DL for HSI-CD. The motivation behind this work is that temporal–spatial–spectral feature extraction between Bi-HSIs is prone to overdetection and underdetection due to hyperspectral imaging being affected by many factors (e.g., atmospheric conditions and sensor jitter). Thus, it is necessary to introduce domain knowledge to guide feature extraction as shown in Fig. 1. In this work, we use the low-rank and sparse representation model to obtain low-frequency low-rank features and high-frequency sparse features from the raw HSI. It is emphasized that the low-rank features contain the intrinsic information of HSI, while the sparse features belong to interference noise such as sunlight. Based on this fundamental consideration, a deep unfolding network based on low-rank and sparse representation (LRSRNet) for HSI-CD is proposed, which includes several key technical steps. First, low-rank and sparse representation is utilized to decouple HSI into low-rank and sparse features. Then, an individually iterative scheme is introduced to update the low-rank features and sparse features. Next, a low-rank and sparse subnetwork (LRSnet) is proposed to learn low-rank estimation, sparse estimation, and hyperspectral reconstruction. Meanwhile, a K-step iterative scheme is applied to LRSnet to update the learnable low-rank and sparse features. Finally, a change detection subnetwork (CDnet) constructed by concise convolution is designed to extract change information from the last updated low-rank features.

The main contributions of this article can be summarized as follows.

1) To the best of our knowledge, low-rank and sparse representation are investigated for the first time using a deep unfolding network for the HSI-CD task, which are a new model–data joint-driven network structure for temporal–spatial–spectral feature capture in Bi-HSIs.
2) Low-rank and sparse feature learning are defined as two independent subproblems. By expanding the iterative optimization update steps into a deep network

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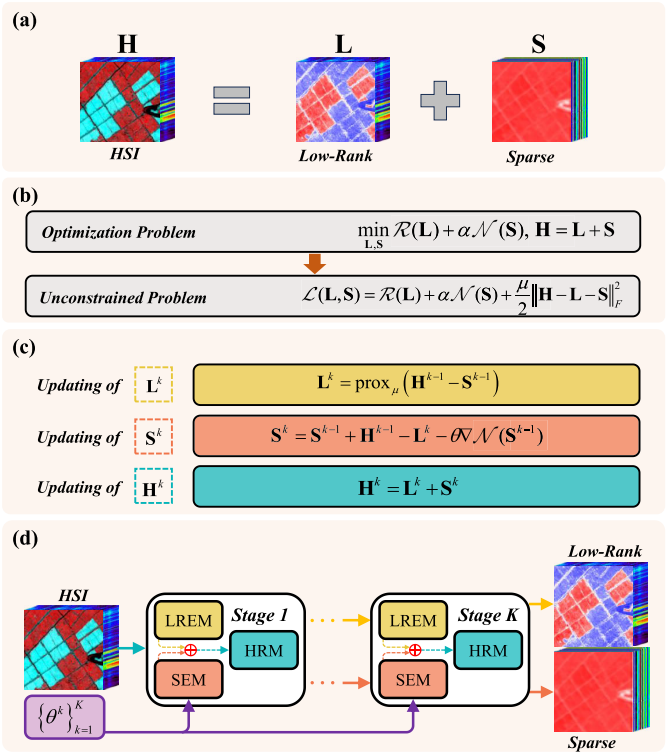


Fig. 1. Fundamental concept of the proposed LRSDNet. Note that the optimization procedure of the LRSD model is unfolded as a deep network. (a) Low-rank and sparse model. (b) LRSD modeling. (c) Model iterative solving. (d) Unfolding into LRSRNet.

framework, the time-consuming and complex matrix calculations are replaced by theory-guided neural networks.

3) A residual attention learning mechanism (RALM) with coordinate and spectral information considerations is designed in the subproblem solving to fully perceive the semantic information of HSI.

Importantly, experiments on several real Bi-HSIs show the performance superiority of the LRSRNet over nine model-driven, data-driven, and model–data joint-driven HSI-CD approaches in both qualitative and quantitative evaluations.

The remainder of this article is organized as follows. Section II provides a review of LRSR in HSI processing including change detection and other fields. Section III introduces the proposed LRSRNet method in detail. Section IV presents experimental analysis and discussions on Bi-HSIs datasets. Section V concludes the article with some remarks and hints at plausible future research.

II. LRSR in HSI Processing

Due to the high correlation between spectral channels and spatial pixels, HSI has intrinsically low-rank and sparse (LRSR) structures [44]. The LRSR representation methods have confirmed to be powerful tools for HSI processing and are widely used in various HSI fields including denoising [45], classification [46], anomaly detection [47], and change detection [48]. For low-rank matrix recovery (LRMR) theory [49], assume that an observed image $\mathbf{Y}$ can be regarded as the combination of an ideal low-rank image $\mathbf{X}$ and a sparse noise matrix $\mathbf{E}$, which can be formulated as follows:

$$\mathbf{Y}=\mathbf{X}+\mathbf{E}. \tag{1}$$

The low-rank matrix $\mathbf{X}$ and sparse noise matrix $\mathbf{S}$ can be recovered according to robust PCA (RPCA) by solving the following optimization problem [49]:

$$\min \|\mathbf{X}\|_*+\lambda\|\mathbf{E}\|_1,\quad \mathbf{Y}=\mathbf{X}+\mathbf{E} \tag{2}$$

where $\|\cdot\|$ refers to the nuclear norm of a matrix, and λ represents a regularization parameter.

Zhou et al. [50] improved the RPCA model (2) by considering both sparse noise $\mathbf{E}$ and Gaussian random noise $\mathbf{N}$. Their observed model is

$$\mathbf{Y}=\mathbf{X}+\mathbf{E}+\mathbf{N} \tag{3}$$

and the corresponding optimization problem is as follows:

$$\min \|\mathbf{X}\|_*+\lambda\|\mathbf{E}\|_1,\quad \text{s.t.},\ \|\mathbf{Y}-\mathbf{X}-\mathbf{E}\|_{\mathrm{F}}^2\le\delta \tag{4}$$

where δ is a constant related to the standard deviation of random noise $\mathbf{N}$. According to low-rank matrix decomposition, some change detection algorithms have been successfully developed. For instance, Chen and Wang [48] proposed a spectrally–spatially regularized low-rank and sparse decomposition for HSI-CD. Zhang et al. [51] introduced an unsupervised HSI-CD algorithm based on PCA and low-rank prior. Hou et al. [52] developed a low-rank deep feature-based CNN for HSI-CD. It is worth mentioning that more information about LRSR theory and applications can be found in [44].

III. Methodology

This section presents the fundamental concepts, design ideas, and solutions of the proposed LRSRNet method. The framework of the proposed method is shown in Figs. 2 and 3. The proposed LRSRNet method will be introduced in six parts, including problem definition, model iterative solution scheme, deep unfolding network scheme, change detection network, and loss function.

A. Problem Definition

Given an HSI $\mathbf{H}$, it can be separated into low-rank features $\mathbf{L}$ and sparse features $\mathbf{S}$ by the physical model, which are defined as follows:

$$\mathbf{H}=\mathbf{L}+\mathbf{S} \tag{5}$$

where $\mathbf{H}$, $\mathbf{L}$, and $\mathbf{S}\in\mathbb{R}^{M\times N\times C}$. M, N, and C are height, width, and number of channels of the HSI. Note that $\mathbf{L}$ represents the intrinsic characteristics of HSI and $\mathbf{S}$ includes the noise caused by hyperspectral imaging.

According to the theory of low-rank and sparse decomposition [50], [53], the sparse features $\mathbf{S}$ are usually constrained to recover the low-rank features $\mathbf{L}$ that can be paired with the given $\mathbf{H}$, thus transforming the problem into

$$\min_{\mathbf{L},\mathbf{S}} \text{rank}(\mathbf{L})+\alpha\|\mathbf{S}\|_0,\quad \text{s.t. } \mathbf{H}=\mathbf{L}+\mathbf{S} \tag{6}$$

where α stands for a trade-off parameter and $\|\cdot\|_0$ is the l_0-norm recorded as the number of nonzero elements.

Due to low-rank functions and L_0 norms are nonconvex and discontinuous, the solution of (6) is NP-hard. Therefore, many model designs employed nuclear norms and L_1 norms

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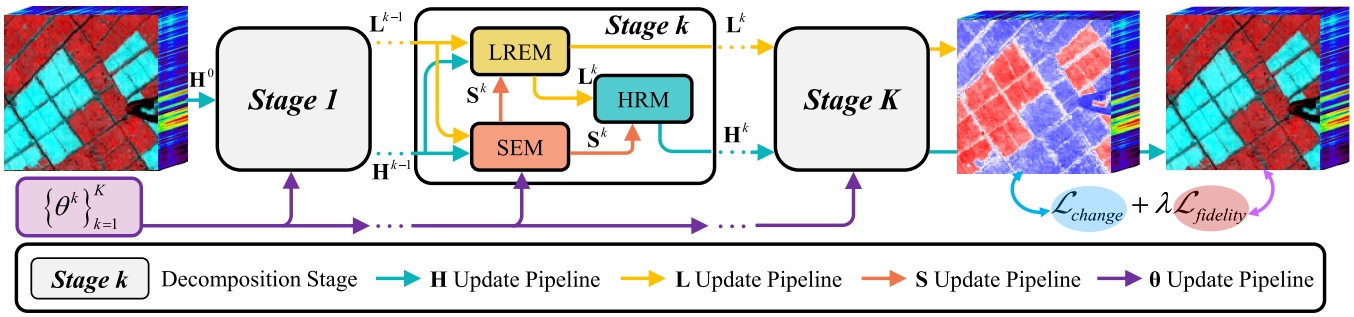


Fig. 2. Overall structure of the proposed LRSRNet for the HSI-CD task.

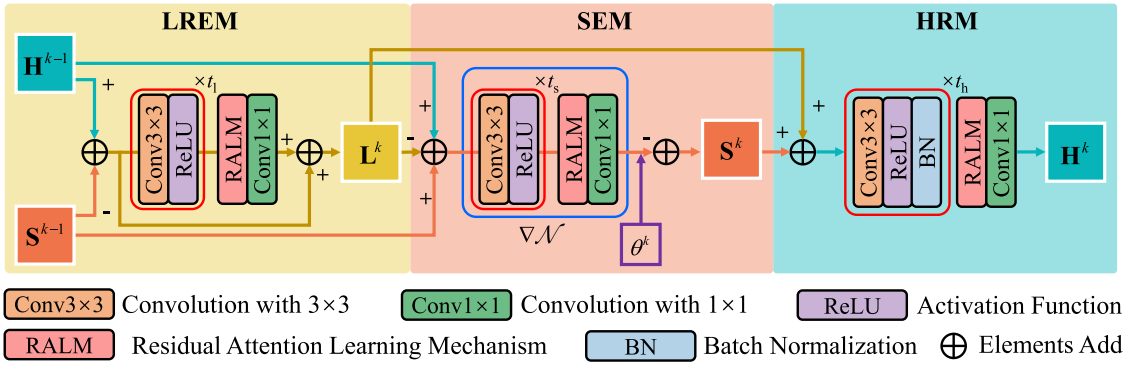


Fig. 3. Detailed network structure of the single stage from Fig. 2 in LRSRNet including LREM, SEM, and HRM.

to replace them according to the principal component pursuit. As such, (6) can be reformulated as follows:

$$\min_{\mathbf{L},\mathbf{S}} \|\mathbf{L}\|_* + \alpha\|\mathbf{S}\|_1, \quad \text{s.t. } \mathbf{H} = \mathbf{L} + \mathbf{S}. \tag{7}$$

In complex hyperspectral scenes, low-frequency low-rank and high-frequency sparseness may differ in complexity and sparsity [44], and an alone norm or rank function is difficult to capture the actual constraints. Thus, $\mathcal{R}(\mathbf{L})$ and $\mathcal{N}(\mathbf{S})$ are introduced to constrain the prior knowledge of low-rank and sparse features, respectively. The expression is as follows:

$$\min_{\mathbf{L},\mathbf{S}} \mathcal{R}(\mathbf{L}) + \alpha\mathcal{N}(\mathbf{S}), \quad \mathbf{H} = \mathbf{L} + \mathbf{S}. \tag{8}$$

Furthermore, in order to avoid the complexity of augmented Lagrange multipliers in the process of updating variables, a more concise and intuitive L_2 norm is utilized to convert the constrained problem to an unconstrained problem [54], the expression is as follows:

$$\mathcal{L}(\mathbf{L},\mathbf{S}) = \mathcal{R}(\mathbf{L}) + \alpha\mathcal{N}(\mathbf{S}) + \frac{\mu}{2}\|\mathbf{H} - \mathbf{L} - \mathbf{S}\|_F^2 \tag{9}$$

where μ represents a penalty coefficient and $\|\cdot\|_F$ refers to the Frobenius norm (F-Norm). We emphasized that the low-rank and sparse features can be individually optimized in an iterative scheme based on (9).

B. Model Iterative Solution Scheme

1) Updating L^*: To update the low-rank components, the subproblem of solving $\mathbf{L}^*$ is formulated as follows:

$$\mathbf{L}^* = \arg\min_{\mathbf{L}} \mathcal{R}(\mathbf{L}) + \frac{\mu}{2}\|\mathbf{L} + \mathbf{S} - \mathbf{H}\|_F^2. \tag{10}$$

According to (7), traditional optimization-based methods often convert $\mathcal{R}(\mathbf{L})$ into $\|\mathbf{L}\|_*$, and then (10) can be degenerated into the sum of the nuclear norm and the L_2 norm. The analytical solution to this problem can be expressed as follows:

$$\mathbf{L}^* = \mathcal{O}_\varepsilon(\mathbf{H} - \mathbf{S}) \tag{11}$$

where $\mathcal{O}_\varepsilon(\cdot)$ refers to the singular value thresholding (SVT) algorithm [55] with threshold of ε. The solution of (11) involves singular value decomposition (SVD) and its functions on singular values. If a neural network is used to replace this process, it is necessary to perform SVD on each neural tensor in the forward propagation, which cannot avoid time-consuming and accuracy issues. To alleviate this problem, an initialization method according to the best rank approximation of SVD is developed in [56] and a technique with various rank constraints on submatrices is investigated in [57]. However, the former still suffers from accuracy issues, while the latter relies on manual tuning and ignores the inherent properties of the image.

Unlike the above works, (11) is degenerated into a constraint function $\mathcal{R}(\mathbf{L})$ instead of employing the nuclear norm and solving it using complex SVD. Importantly, a proximal operator $\text{Prox}_\varepsilon(\cdot)$ is used to approximate the closed-form solution of the low-rank component, which can be defined as follows:

$$\mathbf{L}^* = \text{Prox}_\varepsilon(\mathbf{H} - \mathbf{S}) \tag{12}$$

where the convolutional layer is used to construct $\text{Prox}_\varepsilon(\cdot)$ to solve the optimization problem. We emphasize that the convolutional network can effectively eliminate the time consumption of complex matrix operations. Meanwhile, thanks to

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the nonlinear ability of the network, it can capture the deep semantic features of the image in a data-driven way.

2) Updating S^*: Similarity, to update the sparse components, the subproblem of solving $\mathbf{S}^*$ is expressed as follows:

$$\mathbf{S}^* = \arg\min_{\mathbf{S}} \alpha\mathcal{N}(\mathbf{S}) + \frac{\mu}{2}\|\mathbf{S}+\mathbf{L}-\mathbf{H}\|_F^2. \tag{13}$$

As shown in (7), traditional optimization methods employ the L_1 norm to constrain sparse images, but there are challenges in mapping soft thresholds into neural networks. Besides, the sparse constraints should be shown adaptability to the scene changes. Based on this, the goal of our work is to derive a simple and intuitive representation of the closed-form solution of (13). To solve this problem, the Taylor expansion is applied to $\mathcal{N}(\mathbf{S})$, and then the last $\mathcal{N}(\mathbf{S})$ at $\mathbf{S}^{k-1}$ can be approximated as follows:

$$\hat{\mathcal{N}}\left(\mathbf{S},\mathbf{S}^{k-1}\right) \leftarrow \frac{L_s}{2}\left\|\mathbf{S}-\mathbf{S}^{k-1}+\frac{1}{L_s}\nabla\mathcal{N}\left(\mathbf{S}^{k-1}\right)\right\|_2^2 + C_n \tag{14}$$

$$C_n = -\frac{1}{2L_s}\left\|\nabla\mathcal{N}\left(\mathbf{S}^{k-1}\right)\right\|_2^2 + \mathcal{N}\left(\mathbf{S}^{k-1}\right) \tag{15}$$

where L_s and C_n represent the Lipschitz constant of $\mathcal{N}(\mathbf{S})$ and a constant, respectively. And substituting (14) into (13), the update formulate of $\mathbf{S}$ can be redefined as follows:

$$\begin{aligned}\mathbf{S}^* &= \arg\min_{\mathbf{S}} \alpha\mathcal{N}(\mathbf{S}) + \frac{\mu}{2}\|\mathbf{S}+\mathbf{L}-\mathbf{H}\|_F^2 \\ &= \arg\min_{\mathbf{S}} \frac{L_s}{2}\left\|\mathbf{S}-\mathbf{S}^{k-1}+\frac{1}{L_s}\nabla\mathcal{N}\left(\mathbf{S}^{k-1}\right)\right\|_2^2 \\ &\quad + \frac{\mu}{2}\|\mathbf{S}+\mathbf{L}-\mathbf{H}\|_F^2\end{aligned} \tag{16}$$

which involves only the sum of two L_2 norms instead of the traditional optimization problem with L_1 norm constraints, and thus the solution process does not require help from traditional optimization algorithms or simulated soft-thresholding methods [58]. Succinctly, by taking the derivative of (16) and making its derivative equal to zero, the closed solution of the kth step update $\mathbf{S}$ can be obtained as follows:

$$\begin{aligned}\mathbf{S}^k = &\frac{\alpha L_s}{\alpha L_s+\mu}\mathbf{S}^{k-1} + \frac{\mu}{\alpha L_s+\mu}\left(\mathbf{H}^{k-1}-\mathbf{L}^k\right) \\ &- \frac{\alpha}{\alpha L_s+\mu}\nabla\mathcal{N}\left(\mathbf{S}^{k-1}\right)\end{aligned} \tag{17}$$

where all the three coefficients are constants. By assigning a learnable vector to each vector, (17) for updating the sparse matrix can be rewritten as follows:

$$\begin{aligned}\mathbf{S}^k &= \beta\mathbf{S}^{k-1} + (1-\beta)\left(\mathbf{H}^{k-1}-\mathbf{L}^k\right) - \theta\nabla\mathcal{N}\left(\mathbf{S}^{k-1}\right) \\ \beta &= \frac{\alpha L_s}{\alpha L_s+\mu}, \quad \theta = \frac{\alpha}{\alpha L_s+\mu}.\end{aligned} \tag{18}$$

In our work, an end-to-end deep network satisfying the Lipschitz continuity assumption is employed to learn $\nabla\mathcal{N}(\cdot)$ functions instead of complex matrix operations (e.g., soft-thresholding algorithms). Thus, the reconstructed HSI for the kth update can be obtained as follows:

$$\mathbf{H}^k = \mathbf{L}^k + \mathbf{S}^k. \tag{19}$$

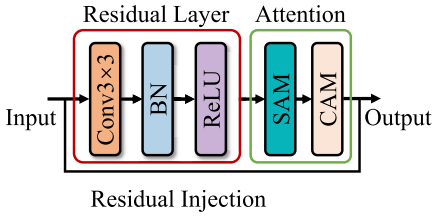


Fig. 4. Overview of the newly designed RALM in the proposed LRSRNet.

C. Deep Unfolding Network Scheme

To reduce complex matrix operations, an LRSnet is introduced to replace the low-rank and sparse decomposition process. As shown in Fig. 2, the input to the network is a single-temporal HSI $\mathbf{I} \in \mathbb{R}^{M\times N\times C}$, where M, N, and C are the height, width, and number of channels of the HSI, respectively, as well as the parameters to be updated are initialized to $\mathbf{H}^0 = \mathbf{I}$ and $\mathbf{S}^0 = 0$. These parameters are then subjected to K decomposition learning stages, each of which corresponds to an iterative low-rank and sparse matrix decomposition process simulating the model-driven approach of multiple iterations of the update operation. Specifically, the updated parameters $\mathbf{H}^{k-1}$ and $\mathbf{S}^{k-1}$ are passed to the kth stage ($k = 1, 2, 3, \ldots, K$). The low-rank component $\mathbf{L}_k$, the sparse component $\mathbf{S}_k$, and the reconstructed $\mathbf{H}_k$ of the current stage are estimated through the low-rank estimation module (LREM), the sparse estimation module (SEM), and the hyperspectral reconstruction module (HRM).

1) Low-Rank Estimation Module: As shown in Fig. 3, an LREM module is proposed to achieve stage learning and estimation of low-rank components in HSI. Inspired by previous deep unfolding network-based works [59], [60], a flexible residual structure $\text{ProxNet}_\varepsilon(\cdot)$ is developed to model the approximation operator $\text{Prox}_\varepsilon(\cdot)$ in (12), which can be formulated as follows:

$$\begin{aligned}\mathbf{L}^k &= \text{ProxNet}_\varepsilon\left(\mathbf{H}^{k-1}-\mathbf{S}^{k-1}\right) \\ &= \mathbf{H}^{k-1}-\mathbf{S}^{k-1}+\mathcal{F}^k\left(\mathbf{H}^{k-1}-\mathbf{S}^{k-1}\right)\end{aligned} \tag{20}$$

where $\mathcal{F}^k(\cdot)$ is denoted as the convolutional group. The stacking of t_l layers is first performed by 3×3 Conv and ReLU, followed by connecting to the proposed RALM module (see Fig. 4) and 1×1 Conv, where ReLU is the activation function. And the RALM is a joint deep feature capture module including convolutional group constructed by 3×3 Conv, ReLU, and BN with L_r layer, followed by the spectral attention mechanism (SAM) and coordinate attention mechanism (CAM). The details of SAM and CAM can be found in [61] and [62], respectively.

2) Sparse Estimation Module: As shown in Fig. 3, the updated low-rank $\mathbf{L}^k$, sparse $\mathbf{S}^{k-1}$, and hyperspectral reconstruction $\mathbf{H}^{k-1}$ are fed as inputs to the SEM. Importantly, β is set to 0.5 to process the three inputs equally, and it is further expressed as follows:

$$\mathbf{S}^k = \mathbf{S}^{k-1} + \mathbf{H}^{k-1} - \mathbf{L}^k - \theta\nabla\mathcal{N}\left(\mathbf{S}^{k-1}\right) \tag{21}$$

where θ is a learnable scalar for the parameter learning task. Note that a is independent, that is, θ^k in each reconstruction stage does not share parameters. Due to the fact that the CNN constructed with convolutional layers and ReLU activation

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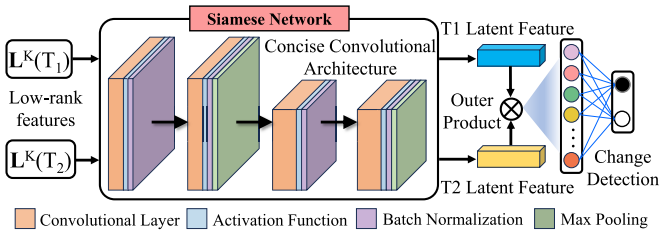


Fig. 5. Outline of Siamese network with simple convolutional architecture for change detection.

functions conforms to the Lipschitz continuum lemma, a CNN architecture with the same structure as the LREM is used to simulate the $\nabla\mathcal{N}(\cdot)$ function. Furthermore, the difference feature between $\mathbf{H}^{k-1}$ and the updated $\mathbf{L}^k$ is utilized to strengthen the sparse features, and the updated representation of $\mathbf{S}^k$ can be defined as follows:

$$\mathbf{S}^k = \mathbf{S}^{k-1} + \mathbf{H}^{k-1} - \mathbf{L}^k - \theta^k \mathcal{Z}^k\left(\mathbf{S}^{k-1}\right) \tag{22}$$

where $\mathcal{Z}^k(\cdot)$ is a CNN with a similar structure to LREM, but its initial module consisting of 3×3 Conv and ReLU is stacked t_s layers.

3) Hyperspectral Reconstruction Module: As shown in Fig. 3, we design the HRM to map the low-rank and sparse features into reconstructed HSI, which is composed of a neural network $\mathcal{J}^k(\cdot)$ that plays the role of going from decomposition task to reconstruction task. The expression is as follows:

$$\mathbf{H}^k = \mathcal{J}^k\left(\mathbf{L}^k + \mathbf{S}^k\right) \tag{23}$$

where $\mathcal{J}^k(\cdot)$ has a similar network structure to $\mathcal{F}^k(\cdot)$ and $\mathcal{Z}^k(\cdot)$. Considering the simplicity and lightness of the architecture, residual learning is removed and introduced into the batch norm (BN) layer. As shown in Fig. 3, the previous convolutional modules are stacked to construct the l_h layer.

D. Change Detection Network

Along with extracting the low-rank and sparse features of the HSIs, we apply the low-rank features containing the intrinsic attributes of the HSIs to the change detection task. As shown in Fig. 5, a concise Siamese change detection subnetwork (CDnet) is proposed for obtaining change information in Bi-HSIs. First, we adopt the 2-D convolution function with 3×3 kernel size, ReLU, BN, and max pooling (MaxPool) as the base unit to gradually capture the latent representation f_{out} of the change information, which can be formulated as follows:

$$\begin{cases} f_{\text{mid}} = \text{BN}\left(\text{ReLU}\left(\text{Conv}_{3\times3}\left(\mathbf{L}^K(T_i)\right)\right)\right) \\ f_{\text{out}} = \text{Maxpool}(\text{BN}(\text{ReLU}(\text{Conv}_{3\times3}(f_{\text{mid}})))) \end{cases} \tag{24}$$

where $\mathbf{L}^K(T_i)$ stands for low-rank feature of HSI at time T_i ($i = 1, 2$). Once the low-rank latent representation of HSI is obtained, the matrix outer product is employed to define the difference features $\mathbf{\Gamma}(d)$ of Bi-HSI, as shown below

$$\mathbf{\Gamma}(d) = \sum_{p\in P} f_{\text{out}}(T_1)^T f_{\text{out}}(T_2) \tag{25}$$

where $\mathbf{\Gamma}(d)$ refers to the difference features of $\mathbf{L}^K(T_1)$ and $\mathbf{L}^K(T_1)$ at position d, and $f_{\text{out}}(T_1)$ and $f_{\text{out}}(T_2)$ are low-rank latent feature at T_1 and T_2 at a location $p \in P$. It is worth emphasizing that compared to performing difference operations on individual pixels [63] to obtain change information, the outer product can better combine neighboring information from patches at different times, rather than subtraction, ratio, or log ratio of corresponding individual pixels.

Next, $\mathbf{\Gamma}(d)$ is reshaped to construct the vectorized feature representation $\mathbf{x}$. Moreover, $\mathbf{x}$ is further processed according to signed square root operation and L_2 normalization to improve the computational performance, which can be denoted as follows:

$$\mathbf{y} = \text{sign}(\mathbf{x})\sqrt{|\mathbf{x}|} \Big/ \left\|\text{sign}(\mathbf{x})\sqrt{|\mathbf{x}|}\right\|_2. \tag{26}$$

where $\text{sign}(\cdot)$ stands for a signed square root operation.

Finally, the standard feature vector $\mathbf{y}$ is fed into the fully connected layer and a softmax function is employed to obtain the output attribute probabilities (i.e., change and nonchange), which can be described as follows:

$$\mathbf{s} \leftrightarrow \text{softmax}(\mathbf{y}) = e^{\mathbf{y}_i} \Big/ \sum_{c=1}^{C} e^{\mathbf{y}_i}. \tag{27}$$

E. Loss Function

In order to facilitate the effective joint learning of LRSnet and CDnet in the HSI-CD task, we design the reconstruction fidelity loss and binary change loss for LRSnet and CDnet, respectively. Specifically, the $\mathcal{L}_1$ loss function is employed to constrain the loss between $\hat{H}$ (i.e., $\mathbf{H}^K$) and $\mathbf{H}$, and the cross-entropy loss function ($\mathcal{L}_{\text{CE}}$) is used to measure the difference between the predicted value of the change and the ground truth (GT), which can be defined as shown below

$$\begin{aligned} \mathcal{L}_{\text{Total}} &= \mathcal{L}_{\text{change}} + \lambda\mathcal{L}_{\text{fidelity}} \\ &= \mathcal{L}_{\text{CE}} + \lambda\mathcal{L}_1 \end{aligned} \tag{28}$$

where λ is a trade-off parameter. Clearly, $\mathcal{L}_{\text{change}}$ is formulated as follows:

$$\mathcal{L}_{\text{CE}} = -\left[\mathbf{y}_{\text{true}} \ln(s(\mathbf{y})) + (1 - \mathbf{y}_{\text{true}}) \ln(s(1 - \mathbf{y}))\right] \tag{29}$$

where $s(\cdot)$ is a softmax function shown in (23) and $\mathbf{y}_{\text{true}}$ is changed GT in Bi-HSIs. $\mathcal{L}_1$ is expressed as follows:

$$\mathcal{L}_1 = \left|\hat{\mathbf{H}} - \mathbf{H}\right|. \tag{30}$$

Note that the specific impact of the value of λ will be discussed in detail later. For the implementation mechanism of LRSRNet in the program, a pseudo code is provided in Algorithm 1.

IV. Experimental Results and Analysis

A. Datasets and Evaluation Metrics

1) Datasets: The first performance validation scenario is the River dataset, which was collected using the Earth Observing-1 (EO-1) Hyperion sensor in a river area in Jiangsu Province, China, on May 3, 2013, and December 31, 2013, respectively, and consists of two HSIs with a total of 242 spectral bands in each image. The single image has a spectral range of 0.4–2.5 μm, spatial and spectral resolutions of 30 m and 10 nm, respectively, and an image size of 463×241 pixels, with the primary change in coverage type in the image being riverine. False-color maps for the T_1 and T_2 HSIs of the river dataset, as well as changed GT, are provided in Fig. 6(a)–(c).

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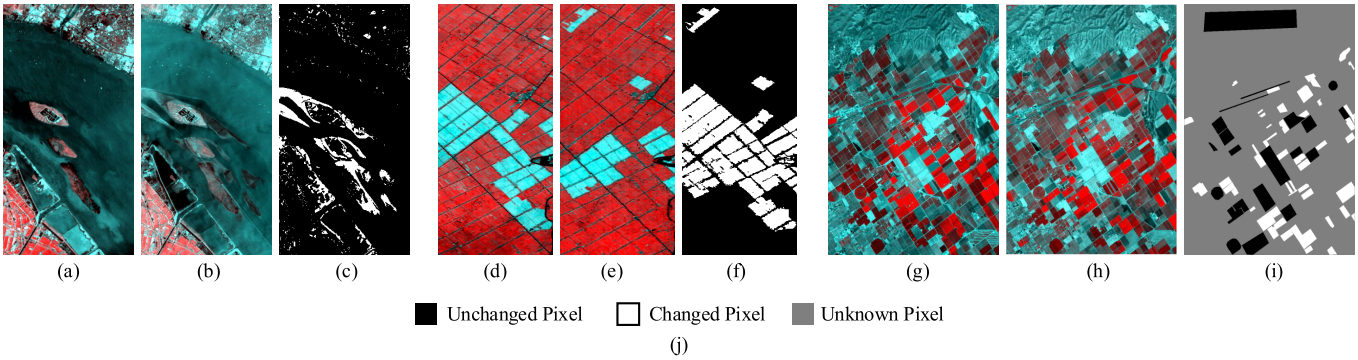


Fig. 6. False color images and GT for all datasets. (a)–(c) False color images of T_1 and T_2 and GT on the River dataset. (d)–(f) False color images of T_1 and T_2 and GT on the Farmland dataset. (g)–(i) False color images of T_1 and T_2 and GT on the Santa Barbara dataset. (j) Pixel legend.

Algorithm 1 LRSRNet for HSI-CD

Input: $\mathbf{H} \in \mathbb{R}^{M\times N\times C}$ from Bi-HSIs
Output: Changed information $\mathbf{y}$ in Bi-HSIs

```
# LRSnet for Bi-HSIs
# Traditional Low-Rank and Sparse Model
H = L + S
L* = arg min_L R(L) + (μ/2)‖L + S − H‖_F^2
S* = arg min_S αN(S) + (μ/2)‖S + L − H‖_F^2
# Its Deep Unfolding Network Scheme
initialization for H^0, S^0 = 0, etc.
# Optimization With Deep Network in Training
while j < epoch do
    for Stage k = 1, 2, ..., K do
        # Low-Rank Estimation Module (LREM)
        L^k = H^{k-1} − S^{k-1} + F^k(H^{k-1} − S^{k-1})
        # Sparse Estimation Module (SEM)
        S^k = S^{k-1} + H^{k-1} − L^k − θ^k Z^k(S^{k-1})
        # Hyperspectral Reconstruction Module (HRM)
        H^k = J^k(L^k + S^k)
    end
    # CDnet for Bi-HSIs
    f_mid = BN(ReLU(Conv_{3×3}(L^K(T_i))))
    f_out = Maxpool(BN(ReLU(Conv_{3×3}(f_mid))))
    Γ(d) = Σ_{p∈P} f_out(T_1)^T f_out(T_2)
    # Calculate L_change and L_fidelity Joint Losses
    L_Total = L_CE + λL_1
end
# Testing Stage for the LRSRNet
Import Bi-HSIs into LRSRNet, and obtain the change
information y through the LRSnet and CDnet.
```

The second validation scenario is the Farmland dataset, which was also acquired by the EO-1 Hyperion sensor on May 3, 2006 and April 23, 2007 at a wetland agricultural area in Yancheng City, Jiangsu Province, China. The spectral range and spectral resolution of the single image are similar to those of the River dataset. The size of this dataset is 420 × 140 pixels and 154 spectral bands after removing the noise and water absorption bands are used for the experiments of performance testing in this work. The main coverage type that changes in HSIs is regular farmland. False color maps of the Farmland dataset along with GT are given in Fig. 6(d)–(f).

The third performance validation scenario is the Santa Barbara dataset, which was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in the urban area of Santa Barbara, California, USA in 2013 and 2014, respectively. The spectral range, spectral resolution, and spatial resolution of the individual images are 0.4–2.5 μm, 10 nm, and 20 m. The dataset images are 984 × 740 pixels in size, contained 224 spectral bands, and the predominantly changing cover type is green vegetation. Fig. 6(g)–(k) presents false color images for T_1 and T_2 HSIs and GT between Bi-HSIs.

2) Evaluation Metrics: To quantify the model performance in change detection, several generalized performance metrics are based on binary confusion matrices, i.e., overall accuracy (OA), kappa coefficient (Kappa), precision (Precision), recall (Recall), and $F1$-score ($F1$). These performance metrics are formulated in detail as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FN}+\mathrm{FP}} \tag{31}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{32}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{33}$$

$$F1 = \frac{2\times(\mathrm{Precision}\times\mathrm{Recall})}{\mathrm{Precision}+\mathrm{Recall}} \tag{34}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA}-P_e}{1-P_e} \tag{35}$$

$$p_e = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})+(\mathrm{TN}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})}{(\mathrm{TP}+\mathrm{FN}+\mathrm{FP}+\mathrm{TN})^2} \tag{36}$$

where TP denotes the number of pixels correctly identified as changing regions, TN denotes the number of pixels correctly identified as unchanging regions, FP denotes the number of pixels incorrectly identified as changing regions, and FN denotes the number of pixels incorrectly identified as unchanging regions (see Table I). The TP, TN, FP, and FN pixels are marked with white, black, red, and green color in the visualization results. The values of TP and TN have larger values indicating better detection performance of the model, while the opposite is true for FP and FN.

B. Competitor Methods and Implementation Details

In order to objectively evaluate the advancement of the proposed LRSRNet method in the HSI-CD task, we compared

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TABLE I
CONFUSION MATRIX FOR BINARY CLASSIFICATION. NOTE THAT TP, FN, FP, AND TN ARE TRUE POSITIVE, FALSE NEGATIVE, FALSE POSITIVE, AND TRUE NEGATIVE, RESPECTIVELY

Confusion Matrix		Prediction	
		Positive (Changed)	Negative (Unchanged)
GT	Positive	TP	FN
	Negative	FP	TN

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR VARIOUS DATASETS

Dataset	Training Set			Testing Set		
	UCH	CH	Total	UCH	CH	Total
River	2400	1200	3600	99485	8498	107983
Farmland	1342	549	1891	43381	17728	61109
Santa Barbara	860	251	1111	79558	51883	131441

• UCH and CH are unchanged and changed samples, respectively.

and analyzed the LRSRNet method with the well-known and state-of-the-art CD methods (including four model-driven, four data-driven, and one model–data joint-driven methods) on the above-mentioned Bi-HSIs data from 3 different scenarios.

1) *Model-Driven Methods:* The CVA [9], k-means with PCA (PCAKM) [11], IR-MAD [12], and SVM [14] are added into our experimental comparison scheme. CVA is a commonly used CD method that provides the intensity and direction of changes in Bi-HSIs. PCAKM is a binary classification result of difference features of Bi-HSIs using k-means. The basic idea of IR-MAD is to weight the iterations based on the cardinal distances on the basis of MAD. SVM is a radial basis function that is used as a kernel function and utilizes a fivefold cross-validation to determine its parameters.
2) *Data-Driven Methods:* The cross-temporal interaction symmetric attention network (CSANet) [64], multiscale diff-changed feature fusion network (MSDFFN) [18], recurrent CNN (ReCNN) [21], and diff-feature contrast enhancement network (DCENet) [65] are adopted as the competitive CD methods in this work. CSANet aims at extracting and fusing the joint spatial–spectral–temporal features of HSI and enhancing the discrimination of changes. MSDFFN is concerned with learning the components of variation in the refinement between Bi-HSIs at different scales to improve feature representation. ReCNN is a combination of RNN and CNN used to learn joint spatial–spectral–temporal features of change features. DCENet combines spectral–spatial feature fusion with Siamese network learning to capture change information in Bi-HSIs.
3) *Joint-Driven Method:* A multiscale convolutional sparse coding (MSCSCNet)-guided deep interpretable network [43] is added into our comparative experiments. MSCSCNet adopts a model-driven network architecture design to capture shared and private sparse coefficients of different scales to determine the variation details between Bi-HSIs.

Moreover, we emphasize that the results of all competing methods that are added to the comparative analysis scheme are reproduced from results in the published literature. In the interest of fairness and objectivity, results are reported as the average of ten replicated experiments using the training and testing configuration in Table II.

In the proposed LRSRNet method, we randomly selected 3% labeled samples from River and Farmland datasets, and 0.2% labeled samples from Santa Barbara dataset as training set. The rest are all used as the test set. Note that the specific quantities can be found in Table II. We utilized the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-4}$ for parameter optimization. An exponential decay strategy is used to update the initial learning rate 2.0×10^{-4} for equal epoch steps, and the number of epochs is set to 200. The batch size used for training on all datasets is 128. The proposed LRSRNet method is trained by the PyTorch framework running in a Windows 11 environment with 32 GB RAM and an NVIDIA RTX 3060 GPU. Note that a detailed hyperparameter analysis will be given later in the Discussion section.

C. Experimental Results

1) Experimental Results on the River Dataset: Table III and Fig. 7 provide the quantitative and qualitative evaluation results of all CD methods (including model-driven, data-driven, and joint-driven) on River dataset. In Table III, it can be seen that CVA, PCAKM, IR-MAD, and SVM, which are the traditional model-driven CD methods, achieve poor performance in terms of Precision and Recall metrics. For instance, CVA achieves 0.9627 in Recall but 0.5444 in Precision; SVM achieves 0.8652 in Precision metrics but only 0.4839 in Recall metrics. For data-driven methods, the CSANet and DECNet methods are able to achieve a better balance than the above-mentioned model-driven methods in terms of underdetection and overdetection, and their OA metrics achieved 0.9551 and 0.9569, respectively. However, it can be also observed that MSDFFN and ReCNN methods are lower than IR-MAD and SVM in terms of OA metrics, which can be attributed to the fact that the labeled samples with a ratio of 3% are difficult to allow MSDFFN and ReCNN to be adequately trained and learned. The joint-driven approaches (i.e., MSCSCNet and LRSRNet) achieve better results in terms of OA, Kappa, and, $F1$ metrics, which means that the joint-driven approaches can effectively integrate the advantages of model-driven and data-driven approaches to improve the HSI-CD performance. In contrast, the proposed LRSRNet method achieves the highest accuracy with limited training samples compared with the MSCSCNet.

Fig. 7 presents the detection results of the GT and each CD method, and the overdetection (FP) and underdetection (FN) are marked with red and green, respectively. CVA maintains a high FP rate, which leads to a high recall rate. SVM has a high FN rate due to its strict judgment of changed features. The PCAKM and IR-MAD methods are similarly varying degrees of overdetection and underdetection. DL-based data-driven methods (i.e., the CSANet, MSDFFN, ReCNN, and DECNet methods) exhibit high FP rate and FN rate due to their sensitivity to the changed features between Bi-HSIs. The detection results of the joint-driven method (e.g., MSCSCNet) are significantly improved compared with the model- and data-driven methods. Compared with all CD methods, the FP rate and FN rate in the detection results of the proposed LRSRNet method in this article are relatively lower than those

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TABLE III
QUALITY METRICS OBTAINED BY VARIOUS HSI-CD METHODS ON RIVER, FARMLAND, AND SANTA BARBARA DATASETS. NOTE THAT BOLD INDICATES THE HIGHEST ACCURACY

Dataset		Different CD methods on the Bi-HSIs datasets.									
		Model-driven				Data-driven				Joint-driven	
		CVA	PCAKM	IR-MAD	SVM	CSANet	MSDFFN	ReCNN	DECNet	MSCSCNet	LRSRNet
River	OA	0.9267	0.9360	0.9490	0.9486	0.9551	0.9403	0.9424	0.9569	0.9625	**0.9648**
	Kappa	0.6575	0.6147	0.6820	0.5955	0.7577	0.7052	0.7011	0.7663	0.7954	**0.8033**
	Precision	0.5444	0.6198	0.7022	**0.8652**	0.6763	0.5972	0.6143	0.6857	0.7122	0.7314
	Recall	**0.9627**	0.6829	0.7178	0.4839	0.9271	0.9616	0.9055	0.9308	0.9545	0.9397
	F1	0.6955	0.6498	0.7099	0.6207	0.7821	0.7368	0.7320	0.7897	0.8157	**0.8225**
Farmland	OA	0.9523	0.9510	0.7765	0.8270	0.9715	0.9627	0.9692	0.9733	0.9732	**0.9744**
	Kappa	0.8855	0.8782	0.3567	0.5765	0.9307	0.9107	0.9251	0.9352	0.9357	**0.9380**
	Precision	0.9024	0.9514	0.7443	0.7074	0.9515	0.9150	0.9500	0.9525	0.9347	**0.9531**
	Recall	0.9370	0.8758	0.3500	0.6881	0.9501	0.9606	0.9436	0.9557	**0.9757**	0.9590
	F1	0.9194	0.9121	0.4761	0.6977	0.9508	0.9372	0.9468	0.9541	0.9548	**0.9560**
Santa Barbara	OA	0.8325	0.7755	0.7408	0.7221	0.9563	0.9629	0.8425	0.9329	0.9760	**0.9788**
	Kappa	0.6519	0.5458	0.3919	0.3815	0.9070	0.9229	0.6655	0.8612	0.9496	**0.9554**
	Precision	0.7741	0.6782	0.9352	0.7194	**0.9895**	0.9346	0.8246	0.8875	0.9769	0.9876
	Recall	0.8107	0.8166	0.3664	0.4809	0.8985	**0.9739**	0.7616	0.9498	0.9617	0.9582
	F1	0.7920	0.7410	0.5265	0.5765	0.9418	0.9538	0.7918	0.9176	0.9693	**0.9727**

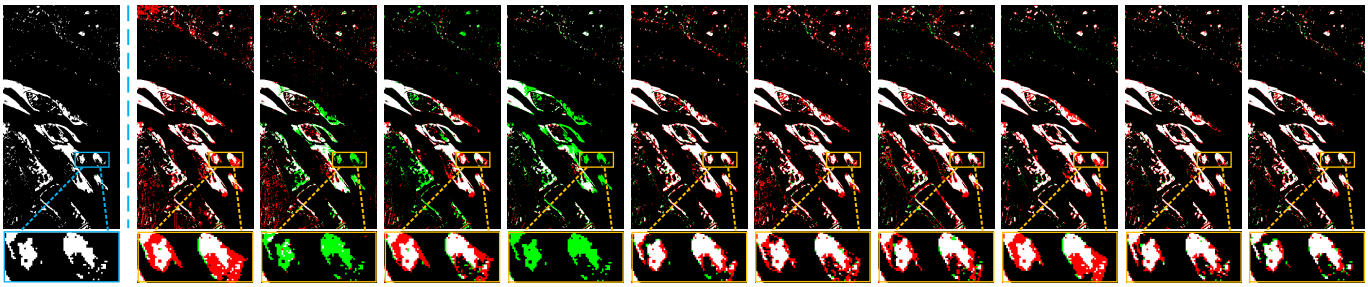


Fig. 7. CD results of various methods on the River dataset. (a) GT. (b) CVA. (c) PCAKM. (d) IR-MAD. (e) SVM. (f) CSANet. (g) MSDFFN. (h) ReCNN. (i) MSCSCNet. (j) LRSRNet. Note that ■ indicates FP (i.e., overdetection) and ■ represents FN (i.e., underdetection).

of competing methods and closer to GT. This means that LRSRNet can achieve better detection performance compared to other CD methods.

2) Experimental Results on the Farmland Dataset: Table III and Fig. 8 present the results of the performance test for each CD method on the Farmland dataset. This dataset contains information on farmland cover changes and noise subject to sensor imaging. It can be observed from Table III that model-driven methods are lower than data-driven methods in terms of five performance evaluation metrics. For instance, the OA metric of CVA is 0.9523, while the OA metrics of CSANet, MSDFFN, ReCNN, and DECNet are 0.9715, 0.9627, 0.9692, and 0.9733, respectively. This suggests that deep network architectures through RNNs and CNNs are able to efficiently extract the information about the differences between Bi-HSIs, which can enhance the CD result effect. The MSCSCNet adopting a joint driving strategy achieves better performance than model- and data-driven methods by designing multiscale sparse coefficient features on the Farmland dataset. The proposed LRSRNet method, as a new joint-driven approach, adopts the strategy of intrinsic feature extraction and interference information separation to capture the real variation features, and thus achieving the highest accuracies in OA, Kappa, Precision, and $F1$, with 0.9744, 0.9344, and 0.9333, respectively. This directly points out the superior performance of the LRSRNet method using low-rank and sparse representation deep features for HSI-CD.

In addition, a visualization of the detection results of each CD method is given in Fig. 8. It can be clearly seen from elliptical window that the model-driven methods exhibit different degrees of overdetection and underdetection, especially the tendency to misclassify the presence of noisy areas as change areas. The data-driven and joint-driven (i.e., MSCSCNet) methods achieve promising results in terms of FP and FN rates, but are prone to overdetection and underdetection in ridge areas. Comparing the model- and data-driven methods, the LRSRNet achieved more promising detection results. The reasons for this are twofold: 1) the use of deep low-rank features to characterize the change information, which enhances the discriminative properties of changed and unchanged pixels; and 2) the interference of noise information avoided by the separation and suppression of sparse features.

3) Experimental Results on the Santa Barbara Dataset: To further evaluate the performance of all CD methods, Bi-HSIs of complex scenarios are used for experiments. Table III and Fig. 9 show the HSI-CD results of model-, data-, and joint-driven methods on the Santa Barbara dataset. It can be seen from Table III that the LRSRNet method achieved the highest accuracy in OA, Kappa, and $F1$ metrics, which means that the LRSRNet still maintains a significant

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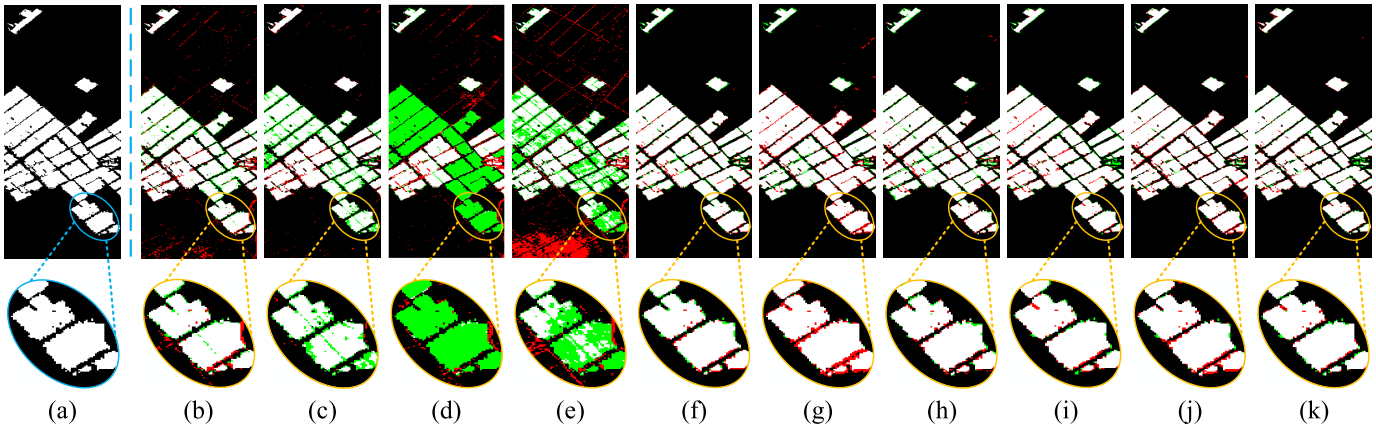


Fig. 8. CD results of various methods on the Farmland dataset. (a) GT. (b) CVA. (c) PCAKM. (d) IR-MAD. (e) SVM. (f) CSANet. (g) MSDFFN. (h) ReCNN. (i) MSCSCNet. (j) LRSRNet. Note that ■ indicates FP (i.e., overdetection) and ■ represents FN (i.e., underdetection).

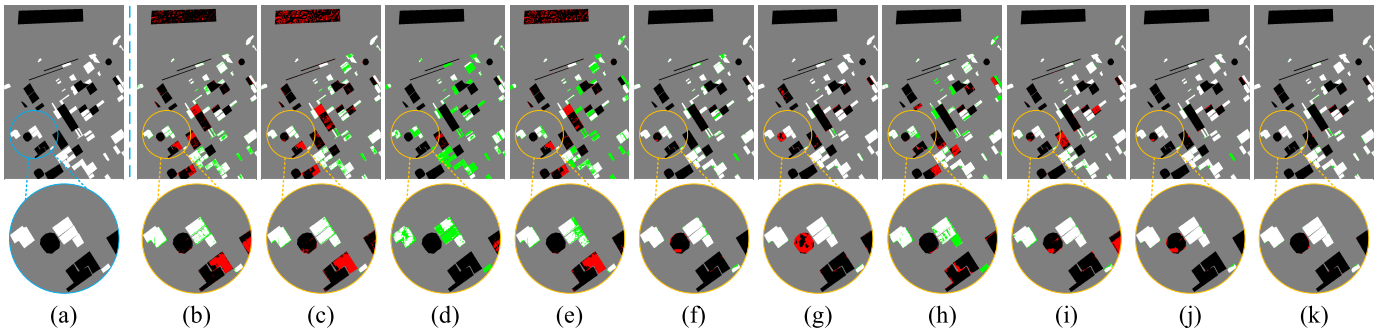


Fig. 9. CD results of various methods on the Santa Barbara dataset. (a) GT. (b) CVA. (c) PCAKM. (d) IR-MAD. (e) SVM. (f) CSANet. (g) MSDFFN. (h) ReCNN. (i) MSCSCNet. (j) LRSRNet. Note that ■ indicates FP (i.e., overdetection) and ■ represents FN (i.e., underdetection).

advantage in the HSI-CD task for complex scenes. However, the LRSRNet achieves suboptimal accuracy in Precision compared to CSANet methods, which is attributed to the imbalance between CSANet methods in overdetection and underdetection, resulting in high Precision metric. Moreover, it can be clearly observed from Fig. 9 (see circular area) that the LRSRNet method achieves significant detection results in terms of overdetection and underdetection compared to all the competing methods (including model-, data-, and joint-driven) in the comparative experimental schedule. This further demonstrates the effectiveness and utility of LRSRNet in the HSI-CD task.

D. Statistical Significance Analysis of the Results

In this section, the McNemar's test [66], [67] is employed to evaluate the statistical significance of the differences between the different HSI-CD methods. The standardized normal test statistic Z can be described as follows:

$$Z = \frac{\mathcal{A}_{12} - \mathcal{A}_{21}}{\sqrt{\mathcal{A}_{12} + \mathcal{A}_{21}}} \tag{37}$$

where $\mathcal{A}_{ij}$ represents the number of samples that are correctly detected by detector i but incorrectly detected by detector j. Z represents the pairwise statistical significance of the difference in change detection between the ith and jth detectors. Note that the test statistic $|Z| \geq 1.96$ indicates that the difference in change detection accuracy between the ith and jth detectors is considered statistically significant at the 5% significance level [66]. The statistical results of the McNemar's test of the LRSRNet and competitive methods are given in Table IV, which demonstrates that the proposed LRSRNet method is superior to model-driven (i.e., CVA, PCAKM, IR-MAD, and SVM), data-driven (i.e., CSANet, MSDFFN, ReCNN, and DECNet), and joint-driven (i.e., MSCSCNet) methods on the River, Farmland, and Santa Barbara. This means that LRSRNet designed with deep unfolding architecture of LRSR is able to improve the change detection accuracy of Bi-HSIs due to the statistically significant test statistics. Meanwhile, this further proves the effectiveness of the proposed LRSRNet method in the HSI-CD task.

TABLE IV
MCNEMAR'S TEST BETWEEN THE PROPOSED METHOD AND OTHER CHANGE DETECTION APPROACHES

Methods	The significance statistic $\|Z\|$		
	River	Farmland	Santa Barbara
LRSRNet vs CVA	47.969	29.458	132.17
LRSRNet vs PCAKM	34.606	31.713	157.97
LRSRNet vs IR-MAD	21.802	107.51	173.96
LRSRNet vs SVM	20.335	90.098	186.51
LRSRNet vs CSANet	15.460	10.753	39.129
LRSRNet vs MSDFFN	34.785	19.608	29.216
LRSRNet vs ReCNN	32.379	13.051	124.33
LRSRNet vs DECNet	13.014	8.321	63.156
LRSRNet vs MSCSCNet	4.057	8.003	5.992

• $|Z| \geq 1.96$ is significant, the larger the value, the more significant.

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TABLE V

ABLATION STUDY FOR THE PROPOSED LRSRNET METHOD ON THE RIVER DATASET. THE BOLD TYPE IS USED FOR THE OPTIMAL INDICATOR, AND THE UNDERLINE IS USED FOR THE SUBOPTIMAL INDICATOR

Ablation stage		LRSD			RALM	Quantitative metrics of performance contribution						
		LREM	SEM	HRM		OA	Kappa	Precision	Recall	F1	Param.	FLOPs
LRSRNet	With	✓	✓	✓	✓	**0.9629**	**0.7934**	**0.7215**	**0.9328**	**0.8137**	4.070M	359.302M
	Without	✗				0.9493	0.7304	0.6483	0.9117	0.7578	3.120M	265.040M
			✗			0.9469	0.7201	0.6363	0.9095	0.7487	3.120M	265.040M
				✗		0.9480	0.7252	0.6412	0.9128	0.7533	3.120M	264.938M
					✗	0.9532	0.7490	0.6668	0.9231	0.7743	3.600M	314.245M
	Baseline	✗	✗	✗	✗	0.9243	0.6214	0.5410	0.8514	0.6616	747.8K	29.302M

Ablation stage		LRSD	RALM			Quantitative metrics of performance contribution						
			RLM	SAM	CAM	OA	Kappa	Precision	Recall	F1	Param.	FLOPs
LRSRNet	With	✓	✓	✓	✓	**0.9629**	**0.7934**	**0.7215**	**0.9328**	**0.8137**	4.070M	359.302M
	Without	✗				0.9444	0.7107	0.6229	0.9130	0.7405	3.599M	314.180M
			✗			0.9539	0.7519	0.6700	0.9245	0.7769	4.042M	358.416M
				✗		0.9544	0.7524	0.6753	0.9151	0.7771	4.062M	359.108M
					✗	0.9575	0.7683	0.6901	0.9275	0.7914	4.048M	358.545M
	Baseline	✗	✗	✗	✗	0.9243	0.6214	0.5410	0.8514	0.6616	747.8K	29.302M

• RLM, SAM, and CAM stand for residual learning mechanism, spectral attention mechanism, and coordinate attention mechanism, respectively.

E. Ablation Analysis

This section conducts ablation experiments on the proposed LRSRNet method on the River dataset to analyze the performance gains of core components. In fact, the LRSRNet method contains two subnetworks, i.e., LRSnet and CDnet. The former aims to capture the deep semantic features of change information from Bi-HSIs, while the latter focuses on the decision of pixel attributes (changed or unchanged). In this experiment, the important modules of LRSD and RALM in LRSnet are ablated one by one to analyze the performance contribution, and CDnet is used as a benchmark model to provide a standard reference. Note that LRSD includes the LREM, SEM, and HRM modules, while RALM includes the RLM, SAM, and CAM modules.

Table V presents the quantitative results of the above ablation experiments on the River dataset. It can be clearly observed that when the LREM, SEM, and HRM modules in LRSD are removed from LRSRNet respectively, its HSI-CD performance degrades significantly. It is worth mentioning that the performance degradation caused by the ablation of the LRSD module is greater than that of the ablation of the RALM module. This means that the three modules in LRSD bring more performance gain to LRSRNet than the RALM module, and also confirms the effectiveness of LRSR-based deep unfolding network in the HSI-CD task. Furthermore, the three modules in RALM are subjected to detailed ablation experiments, and it can be seen that the performance of LRSRNet also degrades as the RLM, SAM, and CAM modules are removed, respectively. This shows that the design of RLM, SAM, and CAM modules in RALM can effectively promote the accuracy of the LRSRNet method in change detection task.

F. Discussion

1) Discussion on the Number of Residual Layers: To investigate the effect of the number of residual layers (i.e., L_r in Section III-B) in RALM on the LRSRNet method in terms of performance and efficiency, we analyzed the detection performance of LRSRNet on three real Bi-HSIs by setting different values for L_r. As shown in Fig. 10, the orange and green rectangular heights in Fig. 10(a)–(c) represent the OA and Kappa metrics, respectively, and the curves are the $F1$ metrics. Fig. 10(d) depicts the OA, Kappa, Param, and FLOPs metrics for L_r at different values. It can be seen from Fig. 10 that the OA, Kappa, and $F1$ metrics vary significantly with the variation of L_r from 1 to 5 and from the overall analysis $L_r = 2$ is a node of turnaround in performance. In addition, as shown in Fig. 10(d), the values of $L_r = 1$ to $L_r = 4$ have a small impact on the detection performance of the LRSRNet, but the number of parameters and FLOPs increase noticeably. Therefore, considering the balance of detection performance over efficiency, $L_r = 2$ is set as the default parameter of the LRSRNet method.

2) Discussion on the Patch Size: The characterization of pixel changes between Bi-HSIs is often represented using a joint representation of the central pixel and its neighborhood information, and thus the patch size depends on the amount of neighborhood information. As shown in Fig. 11, we explored the effect of patch size on the performance of the LRSRNet method in the interval [5 × 5, 7 × 7, 9 × 9, 11 × 11, 13 × 13]. It can be observed that as the patch size is taken from 5 × 5 to 13 × 13, the LRSRNet method shows a trend of degradation in detection performance on the River, Farmland, and Santa Barbara datasets. The reason is that the capture of Bi-HSIs is affected by the acquisition time and imaging sensors, and there are pseudo-change features (e.g., weather) in the larger neighborhood space, thus appropriate neighborhood spectral–spatial information is beneficial for identifying the real change region of Bi-HSIs.

3) Discussion on LRSD Stages: We analyze the performance impact of the important parameter K in LRSnet on the proposed LRSRNet method on the Farmland dataset. We emphasize that the low-rank and sparse components of HSI are extracted more adequately as the value of K increases, but there is an impact on the execution efficiency. Table VI reports the performance metrics of the LRSRNet for different values of K. Focusing on OA and Params, it can be observed that as the value of K increases, the OA metrics show an increasing

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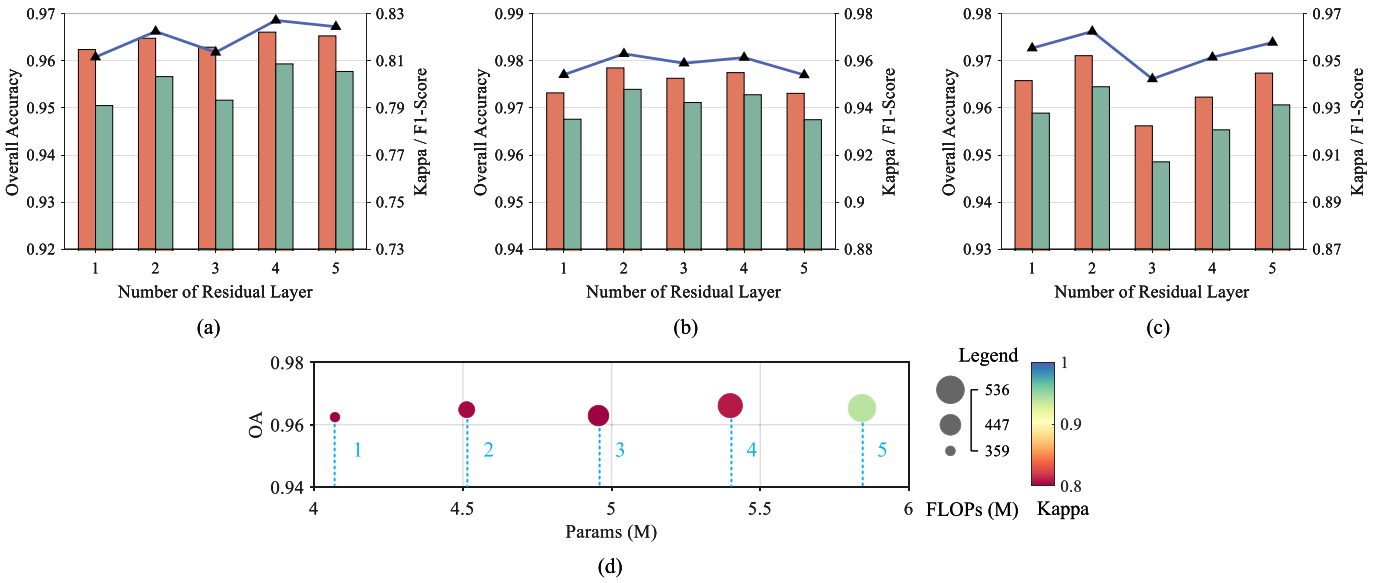


Fig. 10. CD accuracy achieved by the proposed LRSRNet method with different numbers of residual layers on three datasets. (a) River dataset. (b) Farmland dataset. (c) Santa Barbara dataset. (d) Network parameters and FLOPs on (a).

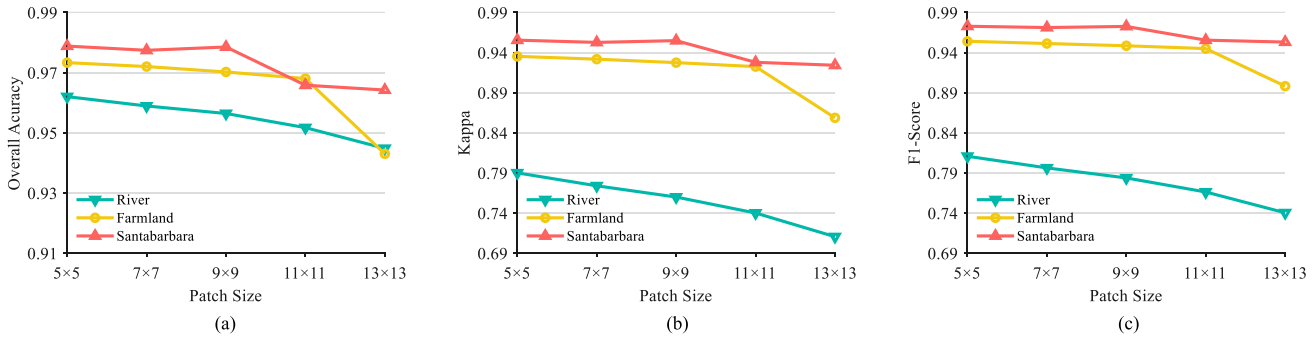


Fig. 11. Effects of different patch sizes on OA, Kappa, and $F1$ score for the proposed LRSRNet method on River, Farmland, and Santa Barbara datasets. (a) OA. (b) Kappa. (c) $F1$ score.

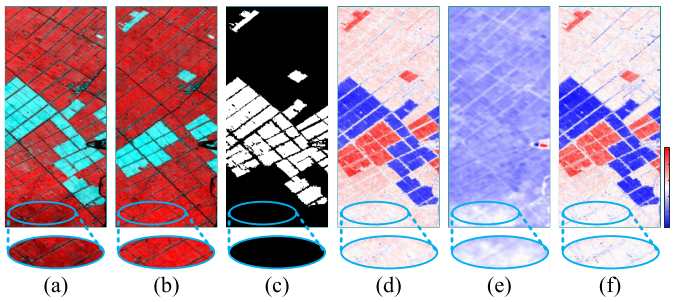


Fig. 12. Visualization of low-rank and sparse features from the LRSD on Farmland dataset. (a) and (b) T_1 and T_2 HSI false color. (c) GT. (d)–(f) First principal component of difference of Bi-HSIs for raw HSI, sparse feature, and low-rank feature.

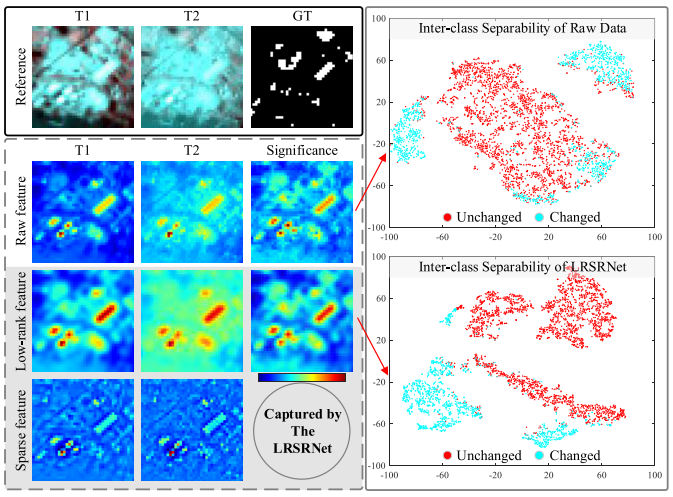


Fig. 13. Feature visualization of the proposed LRSRNet.

and then decreasing trend, and Params shows a sharp increase. Therefore, according to the optimal performance principle, $K = 4$ is adopted as the optimal parameter of LRSnet to LRSRNet.

In addition, we provide quantitative and visualization results for the LRSRNet with and without LRSD process. In Table VII, the LRSRNet method with LRSD obtains better detection results than the LRSRNet method without LRSD. This implies that the low-rank features extracted by the LRSD process can facilitate the identification of change information between Bi-HSIs. Moreover, Fig. 12 presents the results of the visualization of the first principal component of difference of Bi-HSIs for low-rank and sparse features achieved by the LRSRNet, it can be observed from Fig. 12(d)–(f) that

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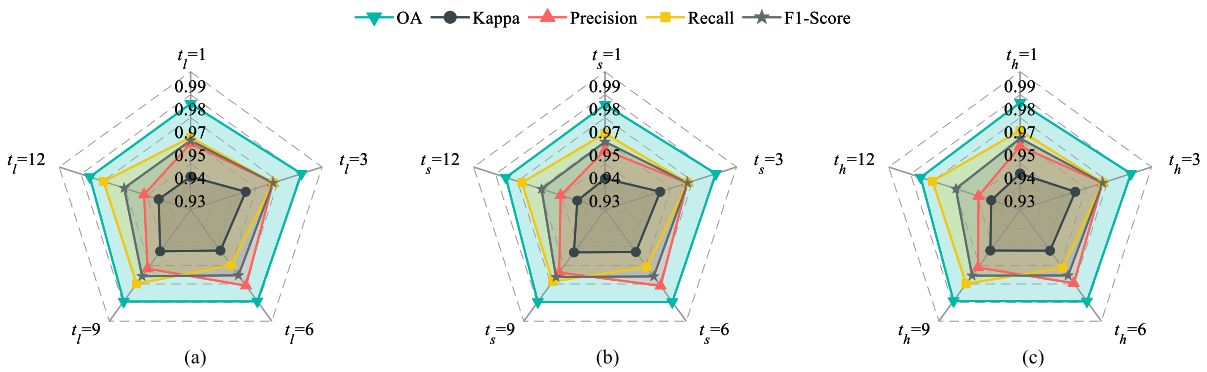


Fig. 14. CD results obtained by the proposed LRSRNet with different layers within sparse learning, low-rank learning, and image recovery stage on the Farmland dataset. (a) Low-rank learning stage t_l. (b) Sparse learning stage t_s. (c) Image recovery stage t_h.

TABLE VI

EFFECT OF STAGE NUMBER K ON THE CHANGED DETECTION PERFORMANCE ON THE FARMLAND DATASET

Stages (K)	CD contribution with LRSD iteration					
	OA	**Kappa**	**Precision**	**Recall**	**F1**	**Params**
K=1	0.9621	0.9071	0.9481	0.9196	0.9336	1.58M
K=2	0.9741	0.9368	0.9663	0.9438	0.9549	2.41M
K=3	0.9742	0.9370	0.9637	0.9467	0.9551	3.24M
K=4	**0.9783**	**0.9474**	**0.9604**	**0.9650**	**0.9627**	4.07M
K=5	0.9778	0.9462	0.9581	0.9656	0.9619	4.90M
K=6	0.9742	0.9375	0.9548	0.9564	0.9556	5.73M
K=7	0.9736	0.9363	0.9465	0.9636	0.9550	6.56M

TABLE VII

CHANGE RESULTS OF THE PROPOSED METHOD WITH OR WITHOUT LRSD ON THE FARMLAND DATASET

Component	CD contribution with or without LRSD				
	OA	**Kappa**	**Precision**	**Recall**	**F1**
without	0.9529	0.7602	0.9394	0.6755	0.7859
with	**0.9783**	**0.9474**	**0.9604**	**0.9650**	**0.9627**

TABLE VIII

CD PERFORMANCE OF THE PROPOSED LRSRNET METHOD WITH VARIOUS LOSS FUNCTIONS ON RIVER, FARMLAND, AND SANTA BARBARA DATASET, RESPECTIVELY

Metrics	River		Farmland		Santa Barbara	
	$\mathcal{L}_{CE}$	$+\mathcal{L}_1$	$\mathcal{L}_{CE}$	$+\mathcal{L}_1$	$\mathcal{L}_{CE}$	$+\mathcal{L}_1$
OA	0.9427	**0.9667**	0.8820	**0.9779**	0.8555	**0.9774**
Kappa	0.7114	**0.8105**	0.6796	**0.9464**	0.6775	**0.9526**
Precision	0.6098	**0.7484**	0.9430	**0.9563**	0.9549	**0.9767**
Recall	0.9471	**0.9285**	0.6171	**0.9679**	0.6425	**0.9656**
F1	0.7419	**0.8288**	0.7522	**0.9621**	0.7776	**0.9711**

noise from raw HSIs can be effectively suppressed to enhance the saliency of changed and unchanged pixels (see elliptical region).

4) Discussion on Feature Visualization: To prove the effectiveness of the proposed LRSRNet method, Fig. 13 provides visual results for LRSRNet. The second row of the left area presents the significance of the raw data T1 and T2 images structure and texture details, as well as the changed pixels of the T1 and T2 images. In contrast, the low-rank features captured by the LRSRNet method can effectively perceive the intrinsic information of the change in ground coverage between T1 and T2 images, and enhance the details of the changed pixels (obtained by LREM). The sparse features captured by the LRSRNet method can obtain the interference information of the changed pixels and suppress the noise expression (obtained by SEM). Furthermore, we visualize the raw data and the discriminative features learned by the LRSRNet method in the right panel (i.e., using t-SNE), and it can be clearly seen that the discriminative features obtained by the LRSRNet method have significant interclass separability compared to the raw data on the River dataset. Therefore, this undoubtedly proves that the proposed LRSRNet method has the ability to guide discriminative feature extraction using low-rank and sparse unfolding networks and its effectiveness in the HSI-CD task.

5) Discussion on LRSnet Stages: LRSnet is an important component of the proposed LRSRNet method. LRSnet consists of three modules, i.e., LREM, SEM, and LRM, where t_l, t_s, and t_h determine the network depth of each of the three modules. In this section, we investigate the effect of t_l, t_s, and t_h on the detection performance of LRSRNet. The value intervals of the t_l, t_s, and t_h parameters are set to [1, 3, 6, 9, 12]. It can be seen from Fig. 14 that the LRSRNet achieves poor detection performance when the above parameter is set to 1, while the detection performance of the LRSRNet has a small fluctuation when the above parameter is gradually set to 3, 6, 9, 12. Moreover, the larger the values of the t_l, t_s, and t_h parameters, the larger the number of parameters of LRSRNet. Therefore, the t_l, t_s, and t_h parameters are all set to 3 based on a consideration of performance and execution efficiency.

6) Discussion on the Loss Function: The loss function of this work consists of two parts, on the one hand, it considers the data fidelity of the reconstructed HSIs output by HRM, and on the other hand, it takes into account the accuracy of the CDnet for the detection of the change information between Bi-HSIs. The data fidelity and detection accuracy are constrained using $\mathcal{L}_1$ and $\mathcal{L}_{\text{CE}}$, respectively. Table VIII reports the performance metrics of LRSRNet under $\mathcal{L}_{\text{CE}}$ working alone and its performance metrics under $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_1$ working jointly on the River, Farmland, and Santa Barbara datasets. It can be observed that the joint $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_1$ loss function is able to drive LRSRNet to achieve promising HSI-CD performance. In addition, we investigated the trade-off parameter λ in the above loss function in Fig. 15, and it can

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TABLE IX
COMPUTATION COST OF VARIOUS METHODS ON RIVER DATASET

Method	Model-driven				Data-driven				Joint-driven	
	CVA	PCAKM	IR-MAD	SVM	CSANet	MSDFFN	ReCNN	DECNet	MSCSCNet	LRSRNet
Params	/	/	/	/	2.4679M	88.771M	878.59K	16.006M	32.224M	4.2992M
FLOPs	/	/	/	/	293.52M	581.97M	5.9034M	520.56M	81.895G	359.30M
Running Time (s)	0.0827	0.1423	0.6997	4.2468	5.7110	6.0349	1.7834	5.5102	150.25	5.9197

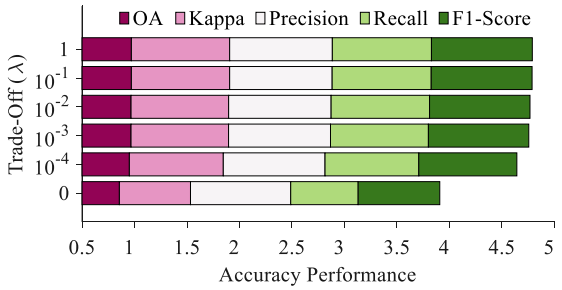


Fig. 15. CD results achieved by the proposed LRSRNet with different trade-off parameters in training stage on the Santa Barbara dataset.

be seen that LRSRNet achieves poor detection performance when $\lambda = 0$, which indicates that the use of $\mathcal{L}_{\mathrm{CE}}$ alone does not enable LRSRNet to learn effectively. The reason for this is that the deep network does not take into account the fidelity of the HSI reconstruction during the training stage, which leads to the imprecise extraction of the low-rank components.

7) Discussion on Computational Cost: This section presents the time consumption of all CD methods in Table IX on the River dataset. Although model-driven algorithms, such as CVA, PCAKM, and IR-MAD, achieve poor detection performance in terms of balance between underdetection and overdetection, these methods have shown high execution efficiency. The CD detection performance of the data-driven method is more advantageous than that of the model-driven method, but it is limited by the number of parameters and FLOPs of the deep network, which takes up more time. The MSCSCNet and LRSRNet belong to the deep network architecture jointly driven by the model and data, aiming to enhance the transparency of the network through model-guided architecture design. The MSCSCNet method achieves better CD accuracy than the above model-driven and data-driven methods. In contrast, the proposed LRSRNet method not only maintains its advantage in CD accuracy, but also achieves acceptable performance in terms of number of parameters, FLOPs, and execution time compared to the advanced MSCSCNet.

V. CONCLUSION

In this article, we propose a new low-rank and sparse representation-based deep unfolding network (LRSRNet) for change detection with Bi-HSIs, which has a distinct feature mining mechanism and a transparent architecture design. We have designed LRSnet and CDnet subnetworks in the proposed LRSRNet architecture. LRSnet is utilized to extract the valuable low-rank information of HSI while suppressing the noise-containing sparse components, and CDnet is employed to discriminate the change information on the low-rank features learned by LRSnet, which fully defines the process and mechanism of feature mining in the HSI-CD task. In addition, we construct a transparent deep network structure by applying the deep unfolding network paradigm to the low-rank and sparse decomposition process to gradually learn and optimize the low-rank and sparse components in HSI. Experiments on three real datasets have demonstrated the superior performance of our LRSRNet method over nine model-driven, data-driven, and model–data joint-driven HSI-CD algorithms.

In the future, we will work on two research lines: 1) integrating feature extraction and change detection into a uniform deep unfolding network to further improve the flexibility of deep network in HSI-CD tasks and 2) to deeply combine the LRSRNet method with the few-shot learning so as to address the challenges of HSI in large-scale and long-time-series cross-modal heterogeneous change detection applications [68].

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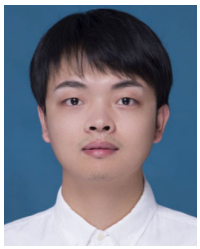
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